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Paleomagnetism of Middle Proterozoic mafic intrusions and Upper Proterozoic (Nankoweap) red beds from the Lower Grand Canyon Supergroup, Arizona

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Abstract

Paleomagnetic data from lavas and dikes of the Unkar igneous suite (16 sites) and sedimentary rocks of the Nankoweap Formation (7 sites), Grand Canyon Supergroup (GCSG), Arizona, provide two primary paleomagnetic poles for Laurentia for the latest Middle Proterozoic (ca. 1090 Ma) at 32°N, 185°E (dp = 6.8°, dm = 9.3°) and early Late Proterozoic (ca. 850–900 Ma) at 10°S, 163°E (dp = 3.5°, dm = 7.0°). A new $^{40}Ar/^{39}Ar$ age determination from an Unkar dike gives an interpreted intrusion age of about 1090 Ma, similar to previously reported geochronologic data for the Cardenas Basalts and associated intrusions. The paleomagnetic data show no evidence of any younger, middle Late Proterozoic tectonothermal event such as has been revealed in previous geochronologic studies of the Unkar igneous suite. The pole position for the Unkar Group Cardenas Basalts and related intrusions is in good agreement with other ca. 1100 Ma paleomagnetic poles from the Keweenawan midcontinent rift deposits and other SW Laurentia diabase intrusions. The close agreement in age and position of the Unkar intrusion (UI) pole with poles derived from rift related rocks from elsewhere in Laurentia indicates that mafic magmatism was essentially synchronous and widespread throughout Laurentia at ca. 1100 Ma, suggesting a large-scale continental magmatic event. The pole position for the Nankoweap Formation, which plots south of the Unkar mafic rocks, is consistent with a younger age of deposition, at about 900 to 850 Ma, than had previously been proposed. Consequently, the inferred ~200 Ma difference in age between the Cardenas Basalts and overlying Nankoweap Formation provides evidence for a third major unconformity within the Grand Canyon sequence.
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1. Introduction

With the increasing popularity of research focused on the proposed Rodinia supercontinent and its subsequent dispersal (Hoffman, 1991; Dalziel, 1992, 1997; Weil et al., 1998; Meert and Powell, 2001; Hartz and Torsvik, 2002), it is crucial to acquire more robust, well-dated paleomagnetic datasets from Precambrian cratons that allow for the creation of more accurately defined paleogeographic reconstructions for the Middle and Late Proterozoic. Reliable paleo-

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magnetic data from Precambrian sedimentary and igneous rocks are vital for reconstructing the tectonic and paleogeographic evolution of the Rodinia supercontinent. Tests of Proterozoic plate reconstructions rely heavily on properly demagnetized rocks that yield well-dated primary paleomagnetic poles, of which there are very few (Buchan et al., 2000). Currently, there are far too many gaps in the paleomagnetic record of most Rodinia cratons to make robust estimates of apparent polar wander path (APWP) segments and, thus, predictions on the details of the relative position of individual cratons and, in particular, on global paleogeography (Powell et al., 1993). The cratons for which reliable data exist typically have incomplete APWPs and/or many individual paleomagnetic results that lack reliable absolute age assignment. To improve on the paleomagnetic database for the Proterozoic of Laurentia, a comprehensive sampling effort has been done throughout the Grand Canyon Supergroup (GCSG), Arizona (Fig. 1).

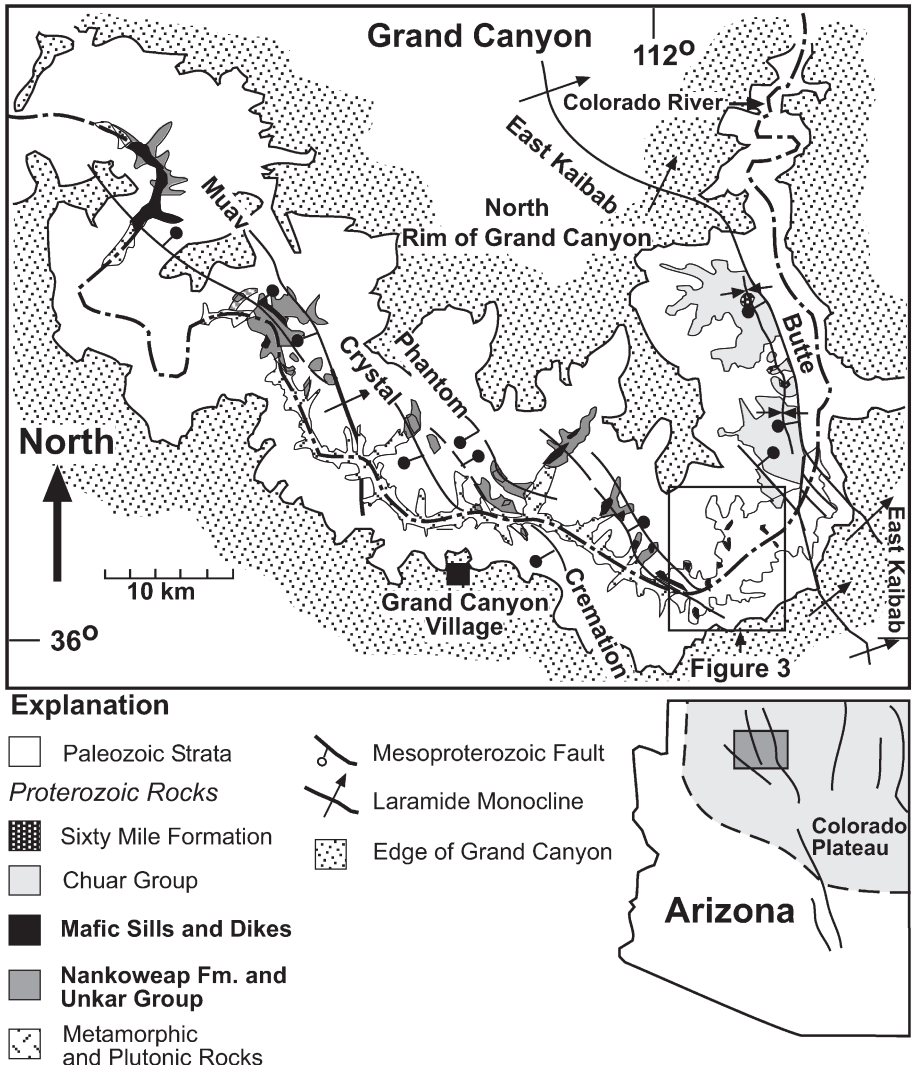


Fig. 1. Simplified geologic map of Precambrian rocks exposed in the Grand Canyon, Arizona. Modified from Timmons et al. (2001).

This study reports new paleomagnetic results from Middle Proterozoic igneous rocks of the lower GCSG and the Upper Proterozoic Nankoweap Formation.

The GCSG is one of the thickest, best preserved and regionally unmetamorphosed Proterozoic sequences in the US for much of the time during which the Rodinia supercontinent existed (1300–750 Ma). The entire GCSG lies unconformably on top of crystalline basement of the Vishnu Schist. The Vishnu Schist was formed during ca. 1.7 Ga metamorphism (Hawkins and Bowring, 1999) and cooled below ~200 °C between 1200 and 1300 Ma (Karlstrom et al., 1997; Timmons et al., 2001). A volcanic ash layer in the Bass limestone near the base of the GCSG is dated at 1253 ± 3 Ma (S. Bowring, unpublished data) giving a basal age of ~1300 Ma. The GCSG is capped unconformably by the Middle Cambrian Tapeats sandstone of the Tonto Group (Fig. 2). The GCSG is divided into two major groups: the Unkar Group (lower GCSG) and the Chuar Group (upper GCSG). The lower GCSG was deposited between 1300 (Bass Limestone) and 1100 Ma (Cardenas Basalts), whereas the upper GCSG was deposited beginning at about 800 Ma during initial Rodinia breakup and Cordillera margin formation (Karlstrom et al., 2000). Sandwiched between the lower and upper GCSG groups is the enigmatic Nankoweap Formation, a thin red bed sequence whose absolute age is yet to be determined (Fig. 2).

The lower GCSG is composed of red sandstones, siltstones and shales, as well as limestone and dolomite layers, and is capped by a 300-m-thick extrusive sequence—the Cardenas Basalts. Cross-cutting lower GCSG strata are numerous mafic intrusions that are proposed to be feeders for the Cardenas Basalts. The most recent isotopic age determinations for the lavas yield a Rb–Sr whole-rock age of 1103 ± 66 Ma (Larson et al., 1994). The petrographic and geochemical relationship between the Cardenas Basalts and mafic dikes has been studied by Hendricks (1989) and Larson et al. (1994), who concluded that the lavas and intrusions were derived from a similar magma source. Initial paleomagnetic studies of these rocks (Elston and Scott, 1973) were promising in that the preserved magnetizations, albeit on the basis of limited demagnetization work, did not resemble those typical of most of the Phanerozoic, suggesting that they had not been remagnetized during younger geologic events.

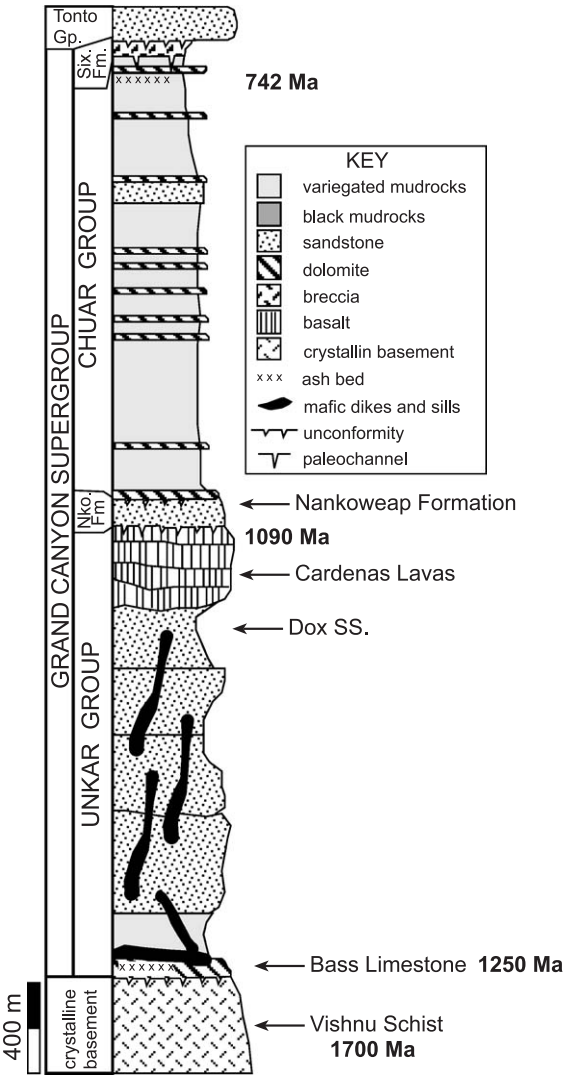


Fig. 2. Simplified stratigraphic section of the Grand Canyon Supergroup, Arizona. Age estimates are provided at the right of the column and are discussed in the text. Column modified from Dehler et al. (2001).

A paleomagnetic, rock magnetic and $^{40}Ar/^{39}Ar$ geochronologic investigation of the mafic igneous rocks within the lower GCSG and the overlying Nankoweap Formation was undertaken to better understand the extent, timing and tectonic significance of large-scale extension within the Rodinia Supercontinent at ~1100 Ma, and to better define Laurentia's late Middle and early Late Proterozoic APWP.

2. Geologic setting

The GCSG, Arizona, records two episodes of latest Middle and Late Proterozoic extension along the ancient Cordilleran margin of Laurentia. The first event, broadly synchronous with Grenville tectonism to the east and southeast (Link et al., 1993; Timmons et al., 2001), is ca. 1250–1100 Ma, and is represented by deposition of Unkar Group strata. The second event, approximately synchronous with rifting of the Rodinia supercontinent, is ca. 800–740 Ma, and is represented by sedimentary rocks of the Chuar Group (Karlstrom et al., 2000). These two major groups are separated by the Nankoweap Formation, which is bound at its top and bottom by slight angular unconformities (Elston and Scott, 1976). However, the duration and importance of the Nankoweap unconformities and, thus, the time interval missing between the two Proterozoic extensional events, is still poorly understood (Elston, 1993; Lucchitta and Hendricks, 1983; Timmons et al., 2001).

The Unkar Group is interpreted as a shallow marine to locally subaerial sequence of lagoonal, tidal flat and deltaic sediments (Hendricks and Stevenson, 1990; Elston, 1989a,b; Link et al., 1993). Deposition of the Unkar Group appears to be discontinuous. The 1253-Ma volcanic ash indicates the Bass Limestone to be about this age. However, $^{40}Ar/^{39}Ar$ dating of detrital muscovite from the basal, Escalante Creek Member of the Dox Sandstone indicates the Dox Sandstone is younger than 1140 Ma (Heizler et al., 1999), but older than the 1103 ± 66 Ma capping Cardenas Basalts. Timmons et al. (2001) attempted to better define the age of the Cardenas Basalts and associated intrusions using the $^{40}Ar/^{39}Ar$ step-heating method. Although the age spectra are disturbed and yield total gas age estimates ranging from about 988 to about 771 Ma, they conclude that the lavas and dikes were emplaced by at least 1050 Ma, consistent with the Rb/Sr data reported by Elston and McKee (1982) and Larson et al. (1994). The disturbance in Cardenas age spectra reported by Timmons et al. (2001) was presumably related to a younger alteration and/or low-temperature heating event (Larson et al., 1994).

The Unkar Group is exposed in full and half grabens within the canyon (Sears, 1990). Field evidence for deposition in an extensional setting includes: dramatic thickness changes in sedimentary units across faults with Proterozoic ancestry, soft sediment deformation, an angular unconformity with the overlying Nankoweap Formation and intraformational faults (Sears, 1990; Timmons et al., 2001). The overall structural features documenting extension within the Unkar Group show a southwest-directed extension direction similar to other proposed Laurentia extensional settings at ca. 1100 Ma (Karlstrom and Humphreys, 1998). The most notable features supporting extension during this time are the mid-continent and Keweenawan rift systems of eastern North America, the Mojave and Arizona terranes and the Apache and Pahrump groups of southwest North America. The geologic setting and available geochronologic data for these regions indicate that mafic magmatism is temporally correlative and was largely confined to the period ca. 1070–1100 Ma (see Harlan, 1993 for a review of available age dates). The cause of this broad tectonomagmatic event affecting much of Laurentia is still debated. Some of the explanations include: (1) Grenville deformation that caused NE–SW extension (Lambeck, 1983; DeRito et al., 1983), (2) intracratonic rifting, (3) mantle insulation (Hoffman, 1989) and (4) southwest North American crust under tectonic compression or an isotropic state of stress at 1.1 Ga (Howard, 1991).

The Cardenas Basalts cap the Unkar Group and were first studied by Powell (1876) and Waltcott (1883). This 300-m-thick sequence of tholeiitic basalt has been subdivided into a lower 100-m-thick coarse-grained, ophitic olivine tholeiite and an upper 200-m-thick fine grained, aphyric quartz tholeiite and tholeiitic andesite (Hendricks, 1989; Larson et al., 1994). Both sequences were erupted subaerially (Larson et al., 1994), with some evidence of subaqueous eruptions in the lower unit (Elston and Mckee, 1982; Lucchitta and Hendricks, 1983). There are at least 12 mappable flows preserved, but it is unclear how many flows existed in the past as the upper surface of the Cardenas Basalts is truncated by Proterozoic erosion (Larson et al., 1994). Beds of siltstone and sandstone are distributed throughout the Cardenas Basalts (Lucchitta and Hendricks, 1983), suggesting periods of quiescence between eruptive flows. Geochemical and isotopic data from the Cardenas Basalts and associated intrusions cutting Unkar strata indicate a genetic relationship between

these igneous rocks, from a slightly enriched peridotitic mantle source (Hendricks, 1989; Larson et al., 1994). However, no feeder relationships have ever been found that can directly link intrusive and extrusive rocks (Hendricks, 1989). Postemplacement/cooling temperatures of the igneous suite never exceeded temperatures of 250 °C and pressures of 1.5 kbar (Larson et al., 1994).

The Nankoweap Formation is a 113–250-m-thick sequence consisting entirely of hematite-cemented sandstones (Elston and Scott, 1976). The Nankoweap Formation was first identified and described by Van Gundy (1934, 1951), but was not formally classified and isolated as a formation until the mapping of Maxson (1961). This red bed sequence is unconformably bound below by the Unkar Group and above by the Chuar Group. The exact genetic relationship of the Nankoweap to older and younger GCSG strata remains poorly understood (Timmons et al., 2001). Furthermore, the age of the formation is undetermined, and has been proposed to be anywhere from 1100 Ma (Elston and Scott, 1976; Elston, 1989b) and, thus, just slightly younger than the Unkar Group, to as young as about ~900 Ma (Lucchitta and Hendricks, 1983) and, thus, closer to the age of inception of deposition of the Chuar Group. Elston and Scott (1976) proposed that the Nankoweap Formation should be divided into two distinct members, a lower ferruginous member and an upper member. Support for this subdivision was based on preliminary paleomagnetic data from the proposed lower ferruginous member (Elston, 1989a,c) that were interpreted to be similar to preliminary results from the Cardenas Basalts (Elston and Scott, 1973). Moreover, Elston (1989c) argued that the redbeds from the lower member of the Nankoweap Formation have strong lithologic similarities to red beds within the underlying Cardenas Basalts sequence. Lucchitta and Hendricks (1983) argued that the lower ferruginous Nankoweap member is in fact part of the underlying Unkar Group, specifically part of the underlying Cardenas Basalts, and that the upper Nankoweap member records much younger deposition. Field evidence suggests that the entire Nankoweap section was deposited in an active extensional environment as evidenced by intraformational faults and soft-sediment deformation (Elston and Scott, 1973; Elston, 1993; Timmons et al., 2001).

The overlying Chuar Group is composed of about 1700 m of sedimentary rock, and is interpreted as dominantly shallow marine and tidal flat deposits, with intermittent deeper water facies (Dehler et al., 2001). Shale, siltstone, sandstone, limestone and dolomite dominate the Chuar rock types in contrast to the generally red sandstone-dominated Unkar Group sequence. A U–Pb date of 742 ± 6 Ma on an ash layer was reported for the upper part of the Chuar Group by Karlstrom et al. (2000). Field evidence for regional extension during the Chuar Group deposition includes intraformational faults, large-scale asymmetric synclines that tighten with depth, suggesting synchronous folding and deposition (Timmons et al., 2001), and the genetic relationship between synclines (i.e., Chuar syncline) and faults with Proterozoic ancestry (e.g., Butte fault) (Elston and Mckee, 1982). In contrast to the NW structural grain found in the Unkar Group, the Chuar Group is characterized by a NS structural grain. The extensional phase recorded by the Chuar Group is thought to be a manifestation of the Neoproterozoic breakup of Rodinia (Dehler et al., 2001; Karlstrom et al., 2000; Timmons et al., 2001).

3. Sampling and experimental methods

Some 250 hand samples were collected in the fall of 1998 and 1999 at 18 sites from the Unkar mafic igneous suite, 6 sites from host Unkar sediments and 8 sites from the Nankoweap Formation (Fig. 3). Site locations were distributed along the main Grand Canyon corridor; severely weathered and highly faulted areas were avoided where possible. Oriented hand samples were collected in the field using a geologic hammer, chisel and magnetic compass (collection of samples using a portable field drill is no longer allowed in the Grand Canyon). Bedding orientations were measured in the field using a magnetic compass.

Seven to ten standard 2.54-cm-diameter paleomagnetic specimens (one per independently oriented sample) from each site were progressively demagnetized in a thermal demagnetizer and measured in a three-axes cryogenic magnetometer in the field-free room at the University of Michigan's paleomagnetic laboratory. Heating was carried out in 50 °C increments to a

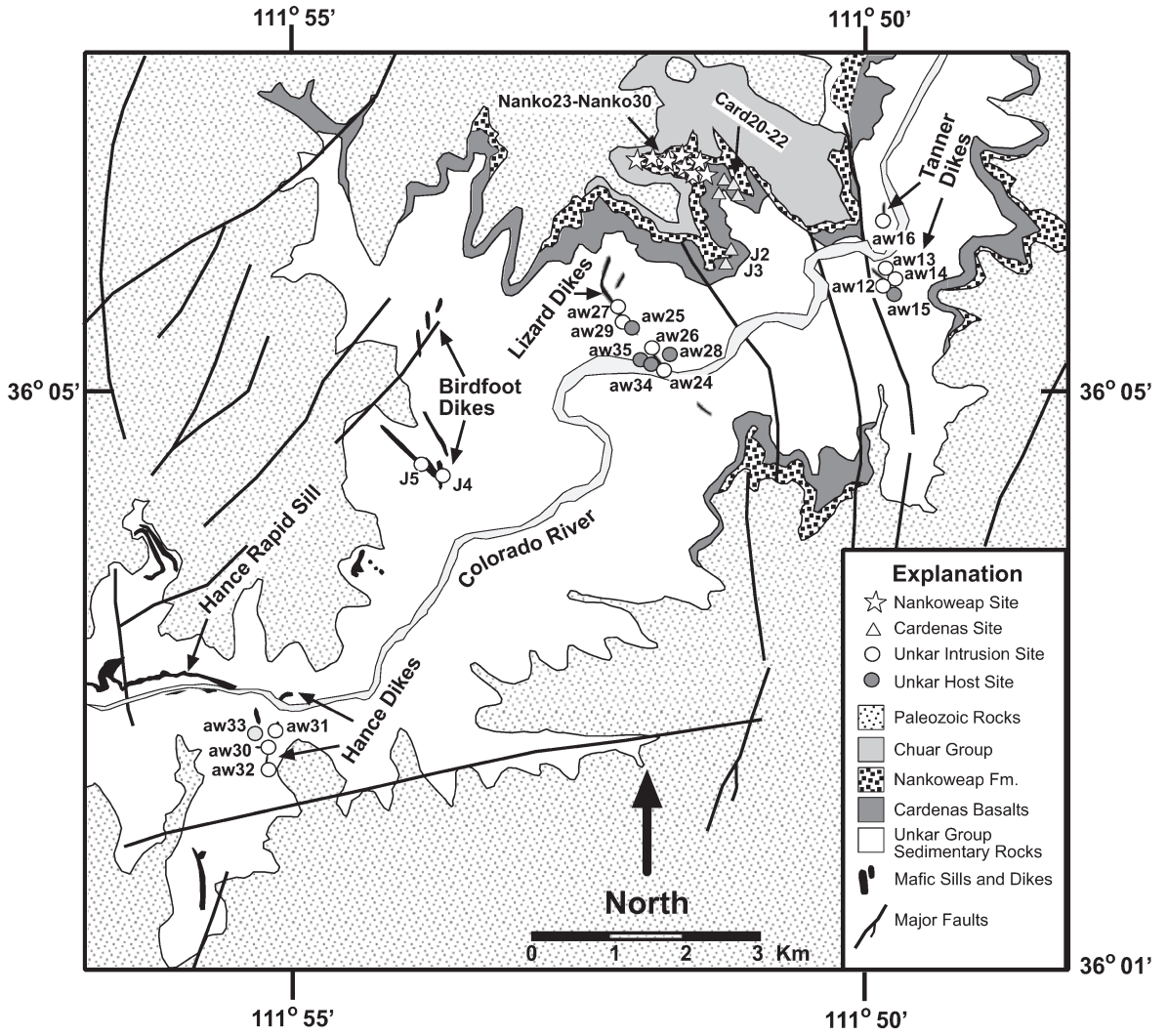


Fig. 3. Geologic map of the main Colorado River corridor through the Grand Canyon, Arizona. Sample site locations highlighted by filled and open symbols. Open circles: Unkar Group intrusions. Grey circles: Unkar Group intrusion contact rock. Grey triangles: Cardenas Basalts. Open stars: Nankoweap Formation. Map modified from Huntoon et al. (1996).

maximum temperature of 680 °C, with reduced increments at temperatures above 350 °C. Alternating field (AF) demagnetization was performed with a Sapphire Instruments SI-4 AF demagnetizer. However, the AF treatment did not adequately and/or fully isolate the principal components of remanence due to the presence of higher coercivity mineral phases (see rock magnetism section). In the thermally treated samples, remanence directions were calculated from principal component analysis (Kirschvink, 1980) of linear demagnetization trajectories identified on paleomagnetic orthogonal projection plots (Zijderveld, 1967). Site-mean directions were calculated by averaging the sample directions, using the method of Fisher (1953).

Where possible, baked contact tests (Everitt and Clegg, 1962) were performed to define the relative age of magnetization acquisition for the mafic intrusions and fold tests were performed to characterize the

relative age of magnetization acquisition by the Nankoweap sediments. At localities where field tests could not be done, comparison of characteristic magnetization directions with field-tested sites was made to attempt to assess the relative timing of magnetization acquisition.

Hysteresis properties of core specimens were measured with a Princeton Vibrating Sample Magnetometer (Micro VSM). Isothermal remanent magnetizations (IRM) were imparted with an electromagnet in fields up to 1.2 T. Finally, thermal demagnetization of three-axes IRM (Lowrie, 1990) was performed on representative samples at the University of New Mexico. For the three-axes measurements, an IRM was first imparted down the long axis of the cores (*z*-axis) with the maximum available field of 3.0 T. Subsequently, 0.3 and 0.03 T fields were applied along the *y* and *x*-axes, respectively.

Two mafic sills (H98-131.6-2, GC9825) collected at river miles 131.6 (Gallaway Canyon) and 109.5 (near the Bass Trail), respectively, were dated by the $^{40}Ar/^{39}Ar$ method. Biotite from H98-131.6-2 was collected from the contact between the sill and the sediments and is interpreted to have grown during sill emplacement. Biotite occurs as a primary phase in GC9825 and was separated directly from the sill. The biotite samples were irradiated at the University of Michigan Ford Reactor in the L67 position for 100 h along with Fish Canyon Sanidine with an assigned age of 27.84 Ma. Following irradiation, the samples were step-heated in a double vacuum Mo resistance furnace and extracted argon was analyzed within an MAP-215-50 mass spectrometer.

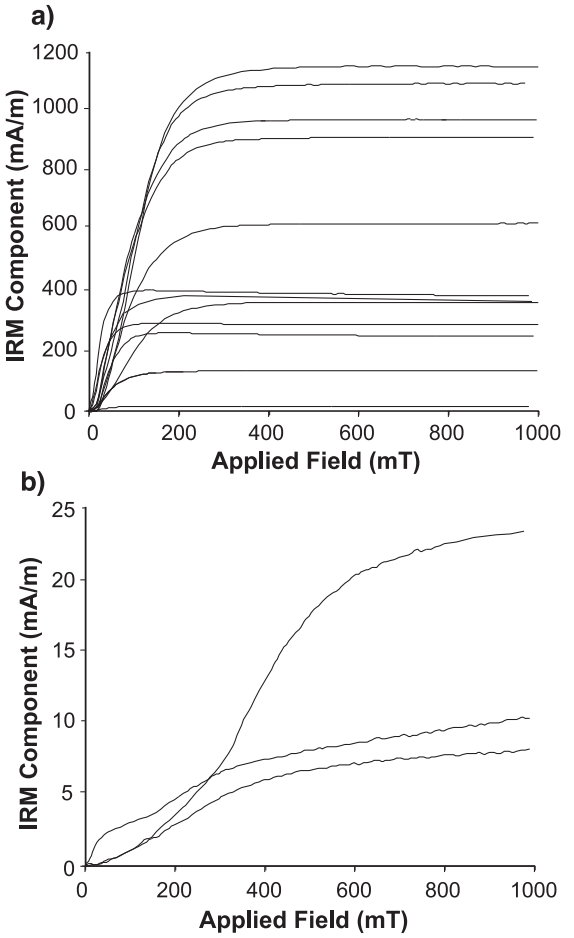


Fig. 4. Typical isothermal remanent magnetization acquisition curves for samples of the Unkar igneous rocks containing (a) only low-coercivity mineral phases and (b) both low- and high-coercivity mineral phases.

4. Results

4.1. Rock magnetism

Several rock magnetic approaches were taken to better understand the overall magnetic mineralogy of the rocks investigated. The documentation of these results, in the context of demonstrating the likelihood that the rocks studied carry ancient and in most cases primary or near-primary magnetizations, is an important facet of our research. IRM measurements were made on core specimens from 15 samples from Unkar Group igneous rocks (Fig. 4). In 12 of the 15 specimens measured, IRM was acquired rapidly in low fields (less than 0.2 T), becoming completely saturated by 0.3 T, indicating a dominance of low- to medium-coercivity mineral phases such as magnetite or titanomagnetite (Fig. 4a). In the remaining three samples of much lower IRM intensities, a higher coercivity mineral phase was present that could not be saturated with the available fields (Fig. 4b). Hematite is the likely candidate for this higher coercivity phase.

Hysteresis measurements were made on 18 core specimens. Several examples of hysteresis loops reveal saturation by 1.0 T and exhibit typical pseudo-

single-domain (PSD), magnetite-dominated unconstricted loops (Fig. 5a and b). Ratios of saturation remanence (Mr) to saturation magnetization (Ms) for those specimens that saturate range from 0.09 to 0.48 (Fig. 5c). Ratios of coercivity of remanence (Hcr) to coercivity (Hc) range from 1.4 to 2.9 (Fig. 5c). With respect to domain state distributions on a Day plot (Day et al., 1977), Mr/Ms and Hcr/Hc values for saturated samples lie in the PSD field for magnetite. The three samples that displayed a higher coercivity phase in IRM acquisition experiments were characterized by unsaturated wasp-waisted hysteresis loops. Furthermore, the wasp-waisted geometry of the loops (Fig. 5d) indicates a bimodal distribution of magnetic phases, likely the combination of low- to medium-coercivity magnetite and higher coercivity hematite. The presence of hematite as a remanence carrier is supported by high laboratory unblocking temperatures (up to 680 °C) for many of the Unkar igneous rocks (Figs. 6 and 7). Demagnetization response, described below, shows the contribution of both magnetite and hematite to the total NRM of most samples (Fig. 6a).

To further test the magnetic mineralogy of the igneous samples, thermal demagnetization of three-axes IRM (Lowrie, 1990) (Fig. 8) allows magnetic minerals with contrasting coercivities to be identified

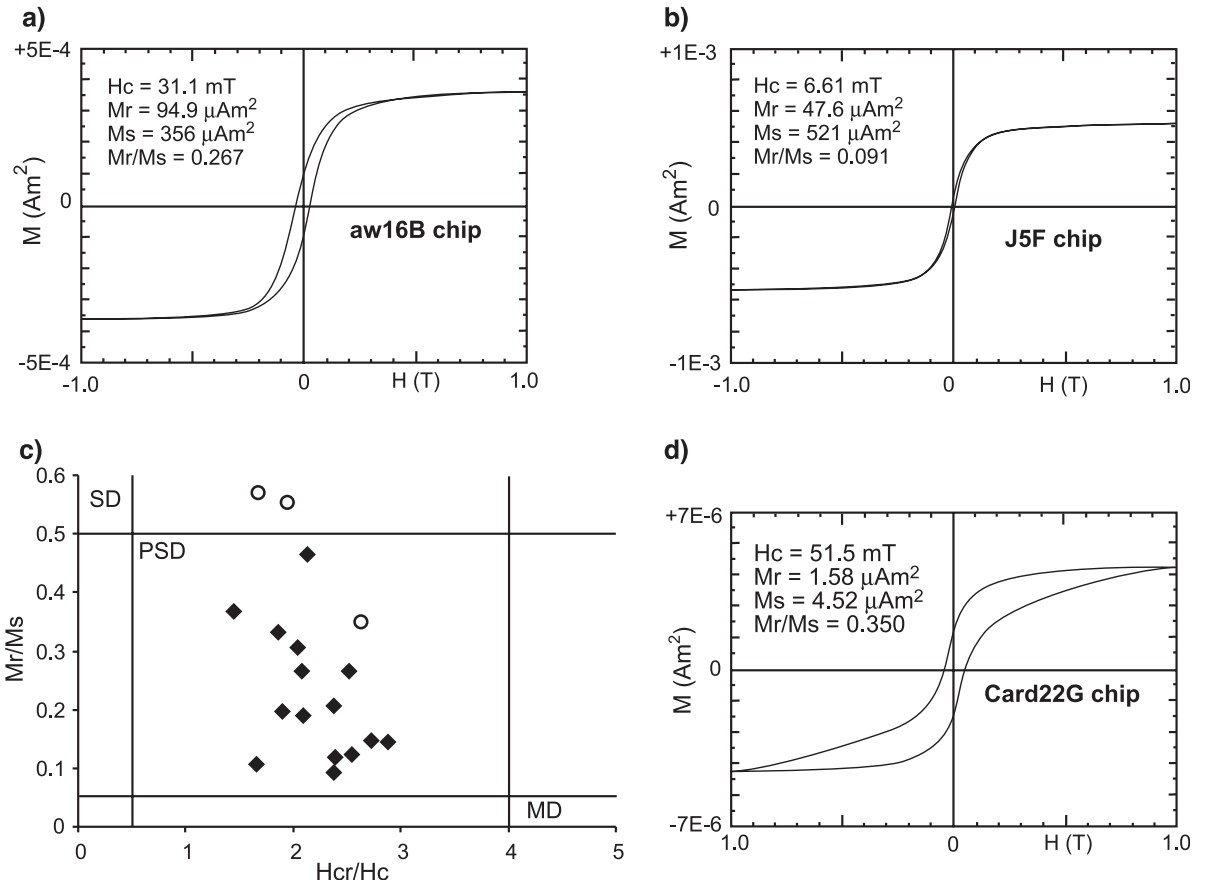


Fig. 5. (a and b) Typical examples of hysteresis loops from Unkar intrusions showing saturation by 1.0 T and typical pseudo-single-domain (PSD), magnetite-dominated unconstricted loops. (c) Hysteresis ratios plotted according to Day et al. (1977) for Unkar intrusions. Filled diamonds distinguishes the samples that were characterized by saturated hysteresis loops (low-coercivity magnetite), and open circles distinguished the samples that had unsaturated wasp-waisted hysteresis loops (combination of low-coercivity magnetite and high-coercivity hematite). (d) Example of a hysteresis loop from Unkar intrusions showing an unsaturated wasp-waisted hysteresis loop typical for mixtures of low-coercivity magnetite and high-coercivity hematite.

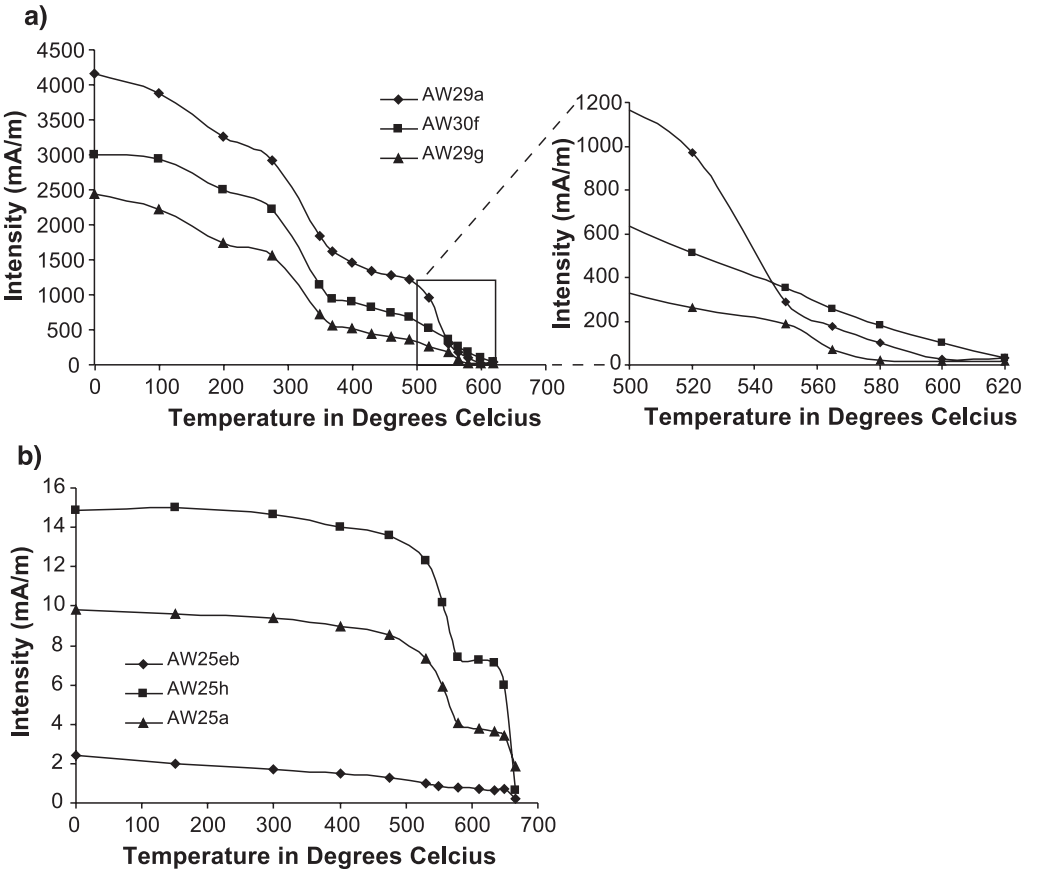


Fig. 6. Thermal demagnetization intensity versus temperature plots of (a) Unkar intrusions and (b) Dox Formation contact rock.

according to their respective unblocking temperatures. Five representative samples were analyzed. The IRM intensities of these samples, even along the *z* component (3.0 T applied field), are all above 100 A/m, and in several cases at least an order of magnitude higher. The dominant magnetic mineral present is activated in 0.3 T and the IRM acquired is largely unblocked below about 580 °C (Fig. 8). We interpret the phase that unblocks below about 580 °C to be nearly pure magnetite. Some samples (in particular J4M, AW14F, and Card22B) show evidence of the discrete unblocking of a second phase at temperatures near or below 400 °C. This phenomenon may be caused by a mixture of coarse maghemite and/or pyrrhotite. Above about 600 °C, the magnetization along the *x* component is essentially zero, while that along the *y* and in particular *z* component smoothly unblocks up to about 675 °C, indicating the relative abundance of hematite. Together, hysteresis measurements, IRM acquisition curves, unblocking temperatures of NRM and three-axes IRM indicate that low-titanium magnetite is the dominant magnetization carrier in Unkar igneous rocks with a measurable contribution from the higher coercivity mineral hematite.

4.2. Paleomagnetism

4.2.1. Unkar igneous rocks

Demagnetization behavior of most Unkar intrusions is relatively straightforward. NRM intensities ranged from 40 to 2500 mA/m. Typical demagnetization response shows that most samples carry a univectorial ancient magnetization component with a minor secondary component (Fig. 7). The secondary component, in geographic coordinates, has a north-seeking declination and a moderate positive inclina-

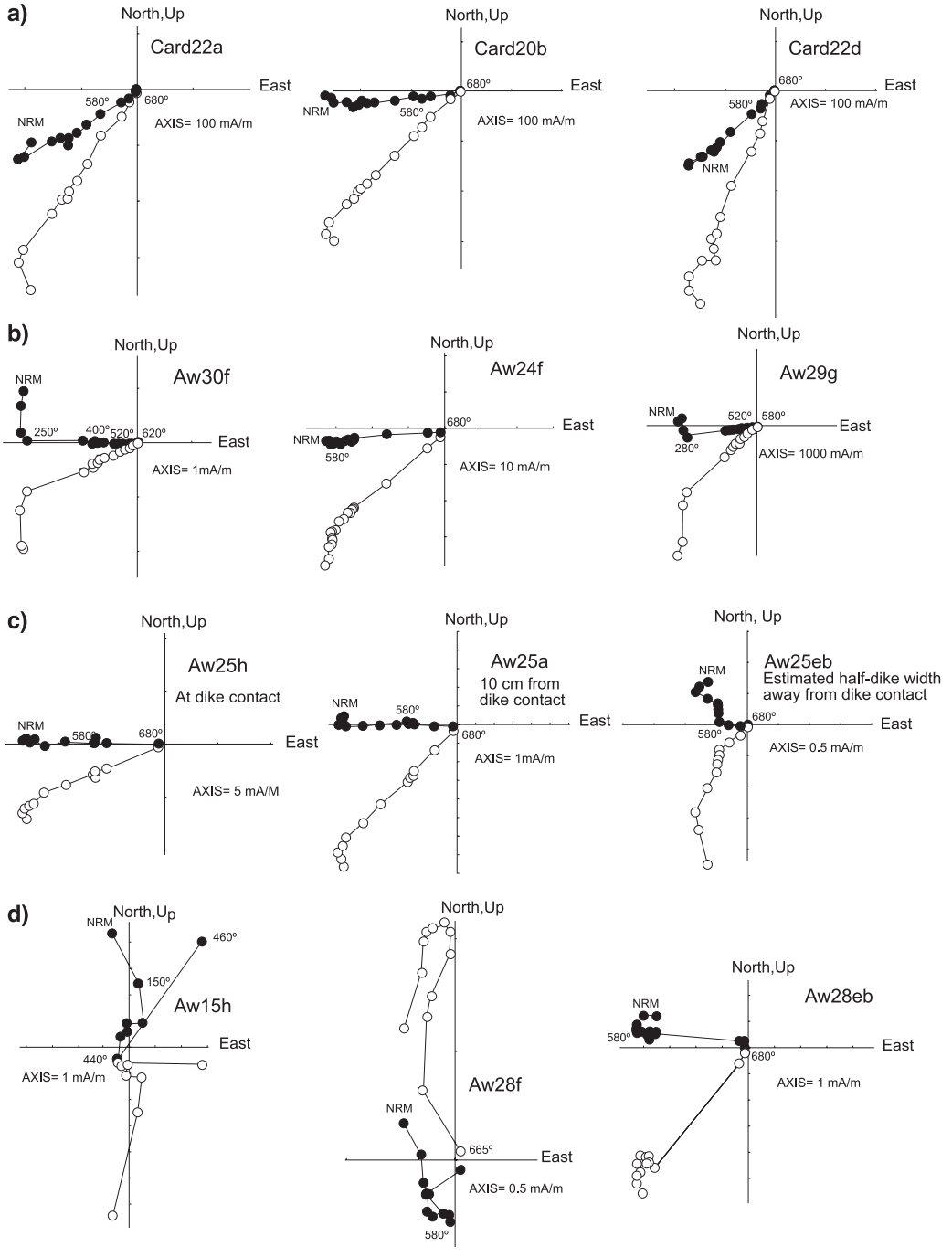


Fig. 7. Demagnetization plots, in geographic coordinates, of representative samples from the Unkar Group: (a) Cardenas Basalts and (b) mafic intrusives. Filled (open) symbols indicate projections onto the horizontal (vertical) plane. (c) Three typical demagnetization plots of representative samples from site aw25 collected along the contract of the Lizard Dike. (d) Three typical demagnetization plots of representative samples from Dox Formation collected at a distance of at least three dike-widths from any observed intrusion.

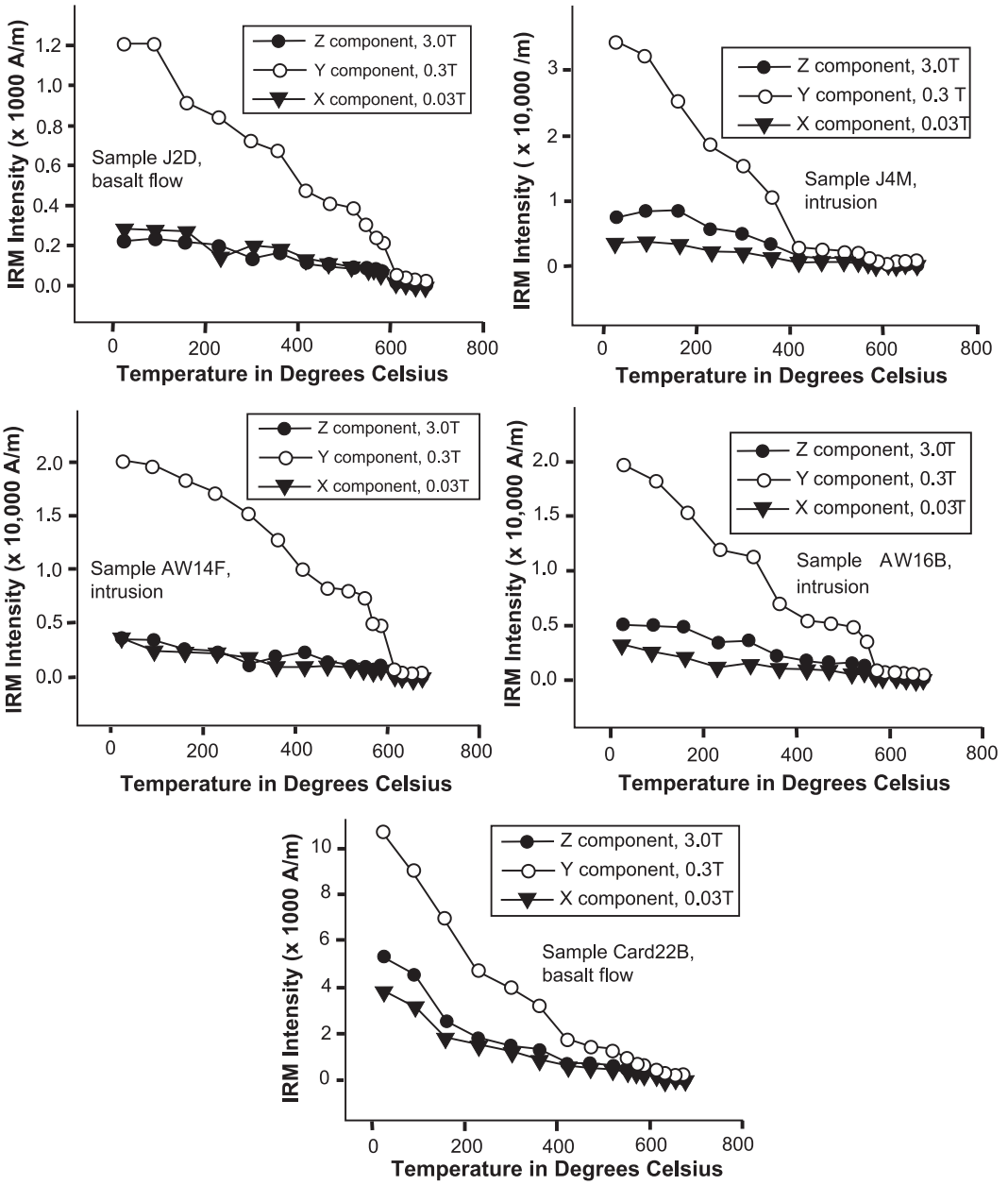


Fig. 8. Thermal demagnetization intensity versus temperature plots of three-component IRM, acquired along the specimen *z*-axis (3.0 T), *y*-axis (0.3 T) and *x*-axis (0.03 T), following the method of Lowrie (1990).

tion, which is subparallel to the Earth's present-day field (PDF) at the sampling location. The PDF component is likely a viscous overprint and is usually removed by 250 °C, with a few exceptions that are not removed until 350 °C.

After removal of the viscous secondary PDF component, most Unkar igneous rocks yield a single component of magnetization that exhibits linear decay to the origin of a demagnetization diagram, which we infer to be the rock's ancient characteristic remanence

Table 1
Paleomagnetic and structural site information and statistical parameters for Unkar Group igneous rocks

Site	N/N_o	Strike	Dip	Dec	Inc	α_{95}	k
Intrusives							
aw12	9/10	310	10	287	61	4	158
aw13	6/8	325	15	276	44	7	97
aw14	10/10	307	12	306	60	5	116
aw16	6/8	307	12	272	56	12	33
aw24	6/8	330	9	254	51	11	38
aw26	8/8	330	9	282	43	7	85
aw27	8/8	330	9	275	49	15	27
aw29	6/8	330	9	267	42	5	132
aw30	8/8	331	15	273	28	9	64
aw31	8/8	331	15	267	47	12	44
aw32	6/8	331	15	310	41	20	16
J4	6/8	303	32	249	26	15	22
J5	8/9	299	30	239	41	11	28
Extrusives							
J2	6/7	295	16	255	42	6	159
J3[a]	–	295	16	–	–	–	–
Card20	6/8	320	25	270	29	14	24
Card21[a]	–	320	25	–	–	–	–
Card22	7/8	320	25	238	49	6	110
Unkar host rock formations (Dox and Hakatai)							
aw15[a,b]	–	310	10	–	–	–	–
aw25[b]	8/8	330	9	271	30	12	26
aw28[a,c]	–	330	9	–	–	–	–
aw33[b]	6/8	331	15	289	45	15	20
aw34[b]	8/8	330	9	270	35	10	35
aw35[a,c]	–	330	9	–	–	–	–

N/N_o is ratio of samples used in the calculation of the site mean direction to the total number of samples collected and measured. Strike/dip, dip direction is 90° clockwise from strike. Dec and Inc are declination and inclination of the in situ site mean directions. α_{95} is semiangle of the cone of confidence about the site mean direction. k is precision parameter (Fisher, 1953).

[a] Site that had no interpretable results.

[b] Site that was collected along intrusion contacts.

[c] Site that was collected at least two dike lengths away from intrusions.

direction (ChRM). This ChRM has a west to west–northwest-seeking declination and a moderate positive inclination (Fig. 7a and b) (Table 1).

All Cardenas Basalt sites were collected from individual flows. Of the five sites collected, only three sites gave interpretable results. The remaining two sites (J3 and Card21) had a combination of unresolvable ChRM directions and/or ChRM directions that were highly dispersed and therefore did not yield an acceptable site mean direction. A likely explanation for the irregular behavior of the two sites is alteration due to geologically recent subaerial weathering. Sites J3 and Card21 were the most ferruginous and visibly altered of the sites collected. The three sites that responded well to demagnetization yielded a univectorial ChRM component of west-seeking declination and a moderate positive inclination. In all cases, the ChRM was isolated between 250 and 680 °C. Demagnetization directions were well defined for individual specimens with maximum angular deviation (MAD) values less than 10°.

The Unkar intrusions sampled range in (exposed) geometry from simple dikes, with widths typically less than 5 m, to more irregular, small dome-shaped features with maximum dimensions measured in tens of meters. All 13 sites from Unkar intrusions gave interpretable results. The ChRM was isolated between 300 and 580 °C with many samples carrying this remanence up to 680 °C. In all cases, the ChRM is of west-seeking declination and a moderate positive inclination (Fig. 7). The fact that both magnetite and hematite contribute as carriers of the ancient remanence in the Unkar intrusions and Cardenas Basalts is interpreted to indicate that the rocks have a primary TRM, residing in low titanium magnetite, and an early acquired chemical magnetization in hematite and possibly maghemite, likely acquired during oxidation of the lava in cooling to ambient temperatures. Oxi-

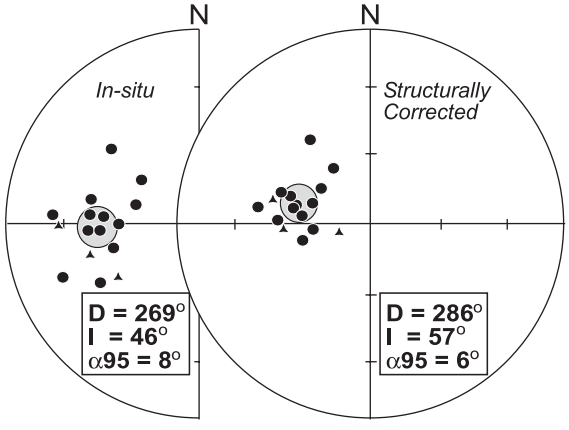


Fig. 9. In situ and structurally corrected paleomagnetic site mean directions (filled symbols, lower hemisphere projection) for the Unkar intrusions (filled circles) and Cardenas Basalts (filled triangles).

dation could have been stimulated by interaction of the igneous rock, at subsolidus conditions, with either a body of water (as has been proposed for parts of the Cardenas Basalts; Elston and Mckee, 1982; Lucchitta and Hendricks, 1983) or with water-saturated sediment (in this case, the Dox Formation). Interestingly, all sites in intrusive rocks collected from the lower parts of the Unkar Group (possibly intruded into more lithified sediment as opposed to water saturated unconsolidated sediment) had magnetizations with laboratory unblocking temperatures below 580 °C. The ChRM directions from those sites in which magnetite

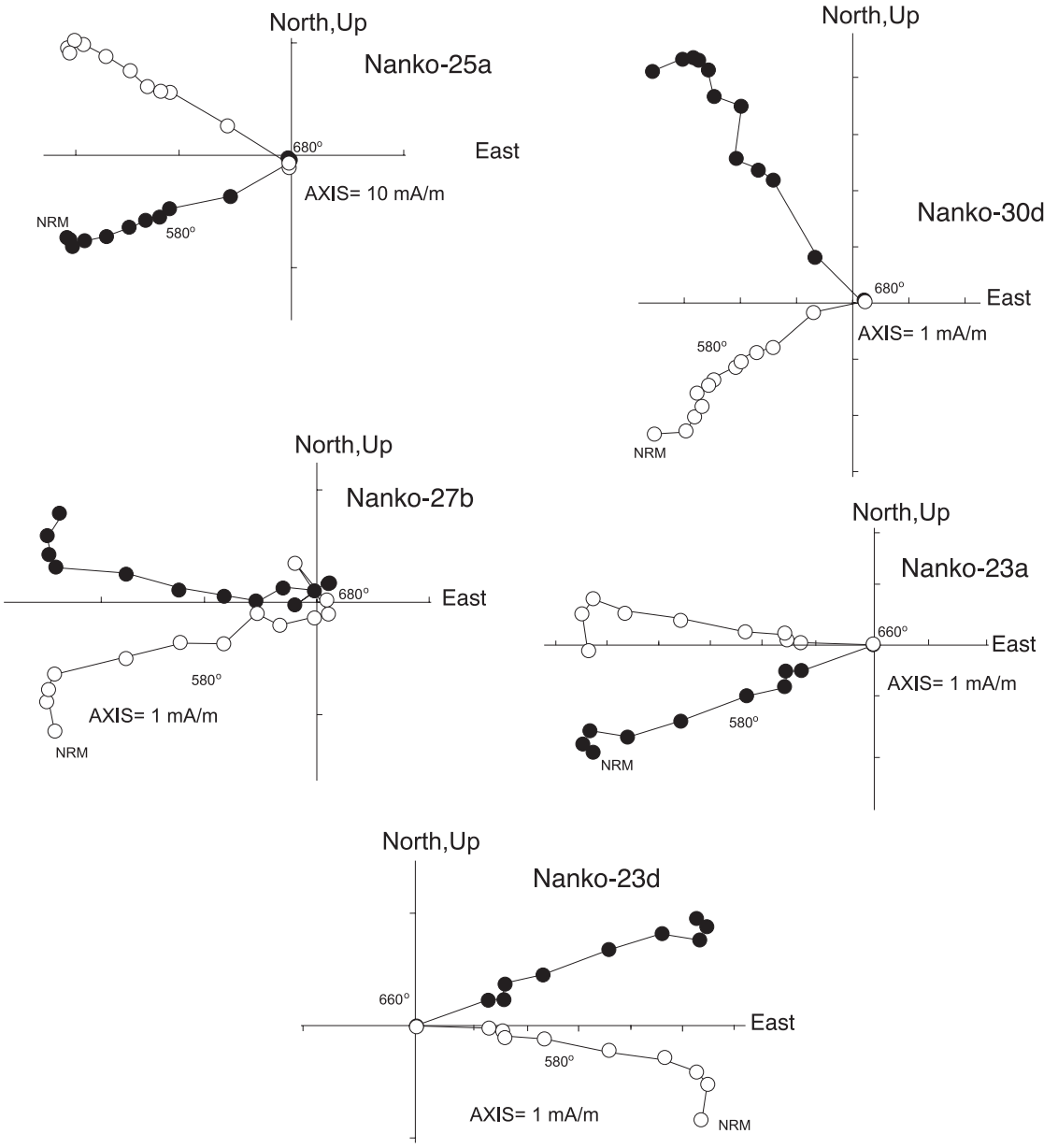


Fig. 10. Typical demagnetization plots of representative samples from the Nankoweap Formation, GCSG, in geographic coordinates. Filled (open) symbols indicate projections onto the horizontal (vertical) plane.

is the principal remanence carrier and those sites in which both magnetite and hematite carry the ancient remanence are statistically indistinguishable (Table 1) and, thus, are not separated for directional analysis.

Several sites were collected for the purpose of performing baked contact tests between Unkar Group strata and the dikes and lavas. One site was collected from the Hakatai Shale (aw33), and five sites were collected from the host Dox Sandstone conformably underlying Cardenas Basalts. Sites aw25 and aw34 were collected along the contact of the Lizard Dike system, sites aw28 and aw35 were collected at least several dike-widths away from the Lizard Dike intrusions, and site aw15 was collected near the Tanner rapid dikes (Fig. 3). Typical demagnetization diagrams are shown in Fig. 7c, and paleomagnetic site information is given in Table 1. At site aw25, samples were collected at incremental distances away from the main Lizard dike. At the dike contact, the Dox samples have the highest magnetization intensities (14.9 mA/m) and the ChRM shows a linear decay to the origin of the demagnetization plot (Fig. 7c) with the ChRM being isolated between 200 and 680 °C (Fig. 6b). At 10 cm away from the dike contact, perpendicular to the strike of the dike, the magnetization is slightly weaker (9.9 mA/m) and is isolated between 250 and 680 °C. At a half-dike width away from the contact, the intensity drops even lower (2.2 mA/m) and the ChRM is isolated between 600 and 680 °C (Fig. 7c). A similar pattern characterized the response from sites aw33 and aw34. Site aw35, collected from the Hakatai Shale, did not give interpretable results. The remaining two sites in Dox strata were collected several dike-widths away from any observed intrusions and were characterized by unstable demagnetization behavior and/or random directions (Fig. 7d). However, two specimens from Dox Formation strata at site aw28 had no obvious spatial association with any intrusions yet carry a ChRM that is well-defined in progressive demagnetization and that is west-seeking and moderately down (Fig. 7d). These specimens likely reflect a primary magnetization that is slightly older than that of the intrusions. The ensemble of results from the contact tests provide a partially positive baked contact result that indicates that the intrusion of the dikes resulted in a secondary thermoremanent magnetization (TRM) in immediately adjacent host Dox Sandstone strata that is subparallel to the ChRM acquired by cross-cutting dikes during initial cooling after emplacement.

Because of the similarity in site-mean directions between the extrusive and intrusive rocks (Fig. 9), the two groups were combined for the purposes of providing a grand mean paleomagnetic direction. After structural corrections for tilt (Fig. 9; Table 1), the total of 16 accepted sites from the Unkar igneous suite gives a mean paleomagnetic direction of $D=285.5°$, $I=57.0°$ ($k=33.9$, $\alpha_{95}=6.4°$, $n=16$) (Fig. 9).

4.2.2. Nankoweap Formation

NRM intensities for Nankoweap Formation red beds range from 2.1 to 32.2 mA/m. Typical demagnetization response shows that almost all samples carry a single univectorial magnetization component with a minor secondary component. The secondary component in in situ coordinates has a north-seeking declination and a moderate positive inclination, which is subparallel to the Earth's present-day field (PDF) at the sampling location. The PDF component is removed by 200 °C and is likely a viscous overprint.

After removal of the viscous PDF component, most Nankoweap Formation specimens yield a single magnetization component that exhibits linear decay to the origin of a demagnetization diagram, which we interpret to be the ancient ChRM of these rocks. This ChRM has a west–southwest declination with a shallow inclination (Fig. 10, Table 2). Several samples

Table 2
Paleomagnetic and structural site information and statistical parameters for the Nankoweap Formation

Site	$N/N_o/N_r$	Strike	Dip	Dec	Inc	α_{95}	k
nanko-23	8/10/2	288	18	250	−11	10	34
nanko-24	7/8/0	150	15	258	7	14	21
nanko-25	7/8/0	303	34	255	−25	7	82
nanko-26	8/8/0	330	28	265	−25	14	16
nanko-27	7/8/0	325	10	269	−16	7	93
nanko-28[a]	–	315	25	–	–	–	–
nanko-29	8/8/2	262	10	263	−2	14	20
nanko-30	3/8/0	215	10	269	−1	14	80

$N/N_o/N_r$ is ratio of samples used in the calculation of the site mean direction to the total number of samples collected and measured and to the total number of opposite polarity specimens measured; strike/dip, dip direction is 90° clockwise from strike. Dec and Inc are declination and inclination of the in situ site mean directions. α_{95} is semiangle of the cone of confidence about the site mean direction. k is precision parameter (Fisher, 1953).

[a] Designates those sites that had no interpretable results.

at some sites carry an oppositely directed magnetization that is east–northeast and shallow down in direction; we note that this behavior is inconsistent at the site level.

Seven of the eight Nankoweap sites provide interpretable results based on stable demagnetization behavior and well-grouped magnetizations. Demagnetization directions were well-defined for individual specimens. Any specimen with a MAD value greater than 10° was not used for directional analysis. The ChRM was isolated between 300 and 680 °C. The rejected site (nanko-28), is located immediately beneath the unconformity between lowermost Chuar Group strata and the Nankoweap Formation. During thermal demagnetization, rocks from this site exhibit unstable, spurious behavior above 300 °C; the results were discarded from further analysis.

Along the traverse of Nankoweap sample sites (Fig. 3), bedding orientations in these strata range in dip direction from northeast to southwest, afford-

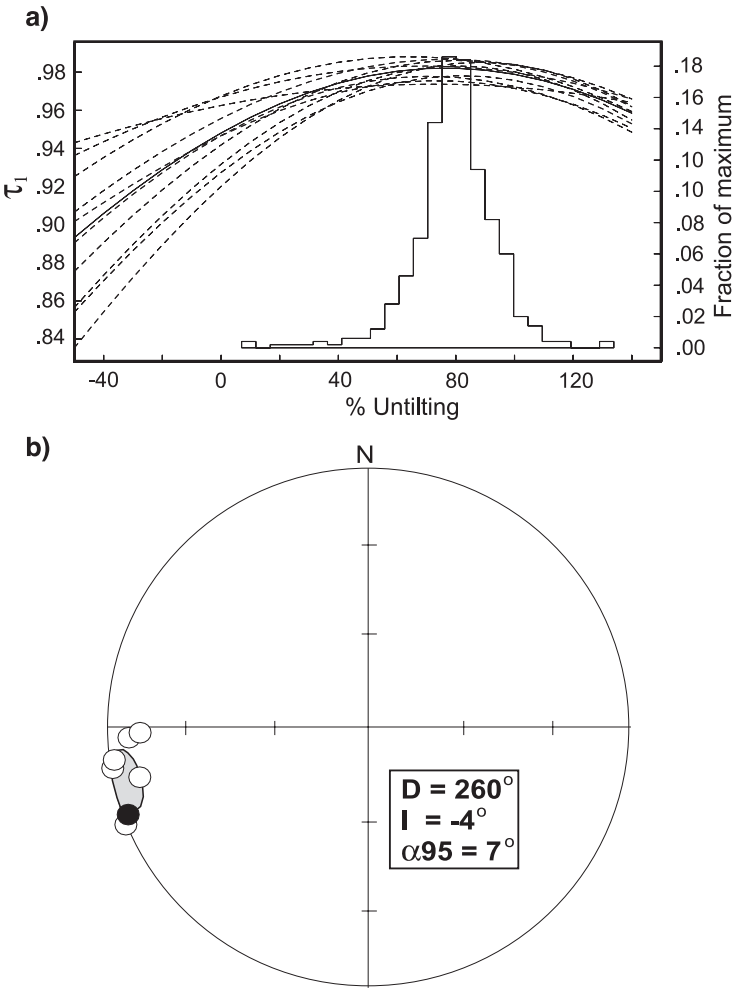


Fig. 11. (a) Incremental parametric bootstrap fold test for the Nankoweap Formation (Tauxe, 1998). The τ_1 parameter represents the largest eigenvalue of the orientation matrices from representative para-data sets. Histogram represents 500 maxima of τ_1 with their respective fraction of maximum given by the right-hand side ordinate axis. Calculated 95% confidence interval is between 59% and 100% unfolding. (b) Structurally corrected paleomagnetic site mean directions for the Nankoweap Formation (filled symbols, lower hemisphere projection; open symbols, upper hemisphere projection).

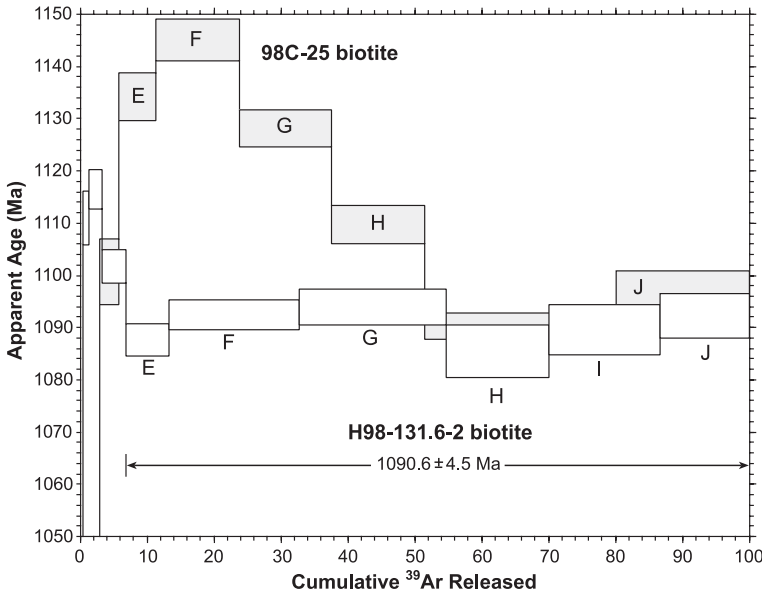


Fig. 12. $^{40}Ar/^{39}Ar$ age spectrum diagrams for biotite from the two mafic sills. The flat spectrum from H98-131.6-2 is interpreted to record emplacement of the sill at 1090.6 ± 4.5 Ma cooling age (2σ error). The final two ca. 1090 Ma steps from the complex spectrum of 98GC-25 are interpreted to also record cooling at about 1090 Ma.

ing a paleomagnetic fold test. The overall fold structure in the Basalt Canyon area plunges gently (about 4°) to the northwest. Therefore, simple bedding-strike tilt corrections were used to rotate the paleomagnetic directions back to their prefolding orientations. Nankoweap Formation site means reach a peak clustering at about 80% unfolding, with a kappa value of 95 (McElhinny, 1964). To test the robustness of the incremental fold test, a parametric bootstrap fold test was applied (Tauxe, 1998), which gives a similar result of maximum clustering at ~80% unfolding and a 95% confidence interval that includes 100% unfolding (Fig. 11a). Given that the Chuar Group unconformably (angular) overlies the Nankoweap Formation and field evidence supports deposition of Nankoweap strata concurrent with deformation, the fold test is interpreted to indicate a prefolding, and very early acquired (primary) magnetization for the Nankoweap Formation. After structural corrections for tilt, the combined seven sites from the Nankoweap Formation give a mean paleomagnetic direction of $D = 259.8°$, $I = -4.1°$ ($k = 72.5$, $\alpha_{95} = 7.1°$, $n = 7$) (Fig. 11b).

4.3. $40Ar/^{39}Ar$ age determination

The two biotite age spectra (Fig. 12, Table 3) are, overall, interpretable although of mixed behavior. Sample H98-131.6-2 yields a generally flat age spectrum with a weighted mean age of 1090.6 ± 4.5 Ma (Fig. 12; Table 3). The steps used for the weighted mean have a slightly elevated mean square weighted deviation (MSWD) value of 2.83 that falls above the 95% confidence level predicted for 5 degrees of freedom (cf. Mahon, 1996). Therefore, the calculated weighted mean uncertainty (Taylor, 1982) has been multiplied by the square root of the MSWD. The 1090.6 ± 4.5 Ma age is interpreted as the intrusion age of this sill (± 2σ error). Biotite from 98GC-25 yields a more complex age spectrum (Fig. 12). Initial steps climb to ~1145 Ma before stepping down to the final two steps that fall between 1090 and 1097 Ma. These final two steps comprise nearly 50% of the total age spectrum and are concordant with the age estimate defined by H98-131.6-2. Despite the complexity of 98GC-25 (presumably caused by excess argon and/or ^{39}Ar recoil) the best age of this sample is interpreted to be about 1090 Ma. This interpretation is

Table 3
Argon isotopic data for sills dated in the Grand Canyon Supergroup

ID	Temperature (°C)	$^{40}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$	$^{36}Ar/^{39}Ar$ ($\times 10^{-3}$)	$^{39}Ar_K$ ($\times 10^{-15}$ mol)	K/Ca	$^{40}Ar^{a}$ (%)	^{39}Ar (%)	Age (Ma)	± 1s (Ma)
98GC-25 biotite, 0.16 mg, J= 0.015774, NM-97, Lab# 9690-01										
A	650	33.63	0.3751	63.90	0.125	1.4	44.1	0.8	380	14
B	750	37.14	0.2673	30.58	0.138	1.9	75.9	1.6	664.6	9.8
C	830	51.32	0.1851	18.43	0.201	2.8	89.5	2.9	984.7	7.7
D	900	55.84	0.1011	9.180	0.466	5.0	95.2	5.8	1100.5	3.2
E	970	56.20	0.0468	2.987	0.912	10.9	98.4	11.4	1134.1	2.3
F	1050	56.49	0.0796	1.556	2.01	6.4	99.2	23.8	1144.9	2.0
G	1120	55.33	0.0653	1.383	2.22	7.8	99.3	37.5	1128.0	1.7
H	1180	54.37	0.0342	2.315	2.25	14.9	98.8	51.5	1109.6	1.8
I	1240	52.83	0.0294	1.230	4.63	17.3	99.3	80.1	1090.1	1.3
J	1350	53.56	0.0257	2.142	3.22	19.9	98.8	100.0	1097.2	1.7
Total gas age			$n=10$		16.2	13.7			1098.6	1.8
No plateau										
H98-131.6-2 biotite, 1.13 mg, J= 0.015763, NM-97, Lab# 9679-01										
A	*650*	*54.33*	*0.1658*	*54.43*	*0.854*	*3.1*	*70.4*	*0.4*	*853.0*	*6.8*
B	*750*	*56.25*	*0.2200*	*8.215*	*2.09*	*2.3*	*95.7*	*1.4*	*1110.9*	*2.6*
C	*830*	*55.36*	*0.1481*	*3.955*	*3.94*	*3.4*	*97.9*	*3.2*	*1116.4*	*1.9*
D	*900*	*53.81*	*0.0408*	*1.878*	*7.62*	*12.5*	*99.0*	*6.8*	*1101.7*	*1.6*
E	970	52.62	0.0391	0.9567	13.9	13.0	99.5	13.3	1087.5	1.6
F	1050	52.74	0.0086	0.2760	41.5	59.0	99.8	32.7	1092.4	1.4
G	1120	52.81	0.0036	0.1770	46.9	140.2	99.9	54.7	1093.9	1.7
H	1180	52.36	0.0081	0.5016	32.9	63.3	99.7	70.1	1085.4	2.5
I	1240	52.59	0.0278	0.4094	35.4	18.4	99.8	86.7	1089.4	2.4
J	1650	53.08	0.1845	1.555	28.5	2.8	99.2	100.0	1092.1	2.2
Total gas age			$n=10$	213.6	56.8				1090.8	1.8
Plateau		MSWD = 2.8	$n=6$	steps E–J	199.1	60.4		93.2	**1090.6**	**4.5**[a]

Isotopic ratios corrected for blank, radioactive decay and mass discrimination not corrected for interfering reactions. Individual analyses show analytical error only. Plateau age error also includes error in J and irradiation parameters. Analyses in italics are excluded from plateau age calculations.

Correction factors: $(^{39}Ar/^{37}Ar)_{Ca} = 0.00070 \pm 0.00005$; $(^{36}Ar/^{37}Ar)_{Ca} = 0.00027 \pm 0.00001$; $(^{40}Ar/^{39}Ar)_{K} = 0.0291 \pm 0.0008$.

Argon extraction system blanks plus mass spectrometer backgrounds: 120, 0.8, 0.4, 0.6, 0.4×10^{-17} for masses 40, 39, 38, 37 and 36, respectively.

[a] 2σ error.

partially based on the apparent concordance of the final two heating steps with the simpler spectrum given by H98-131.6-2.

5. Discussion

The characteristic magnetizations identified in the Unkar igneous suite and Nankoweap Formation provide two distinct ancient paleomagnetic pole positions. The Unkar igneous suite magnetization (UI) is defined by 16 sites from six different intrusions and three different flows of the Cardenas Basalts. The directions for this magnetization are well-defined, of uniform polarity and are of west-seeking declination and moderate positive inclination. A mean paleomagnetic pole, calculated assuming a geocentric axial dipole during the Proterozoic, is located at 32°N, 185°E (dp = 6.8, dm = 9.3) in the north Pacific (Table 4; Fig. 13). We interpret the UI pole to be primary based on a partial contact test with the host Dox Formation strata.

The Nankoweap Formation magnetization (NA) is defined by seven sites along a continuous stratigraphic transect from strata directly overlying the Cardenas Basalts to immediately below the overlying Tanner Member of the Chuar Group. The directions for this magnetization, carried in hematite, are well-defined and have a west–southwest declination and shallow

Table 4
Paleomagnetic pole information for Unkar intrusions and Nankoweap Formation

Pole location	Approximate age (Ma)	N/N_o	Dec	Inc	α_{95}	Plong.	Plat.	dp	dm
Cardenas Basalts/Intrusions	1090	16/18	285.5	57.0	6	185	32	6.8	9.3
Nankoweap Formation	900	7/8	259.8	−4.1	7	163	−10	3.5	7.0

N/N_o is ratio of samples used in the calculation of the site mean direction to the total number of samples collected and measured. Dec and Inc are declination and inclination of the mean paleomagnetic directions. α_{95} is semiangle of the cone of confidence about the mean direction. Plong. and Plat. are the latitude and longitude of the mean paleomagnetic pole. dp/dm are the semiaxes of the cone of 95% confidence about the pole. The value of the paleomagnetic pole in the table is based on the tilt-corrected site mean directions.

negative inclination. A mean paleomagnetic pole, calculated assuming a geocentric axial dipole during the time of magnetization acquisition, is located at 10°S, 163°E (dp = 3.5, dm = 7.0) in the south Pacific (Table 4; Fig. 13). The NA pole is considered primary, or at least acquired soon after deposition, based on a positive fold test, and the available field evidence that suggests that deposition of Nankoweap Formation strata was synchronous with deformation.

The previous paleomagnetic work done throughout the Grand Canyon Supergroup, Arizona, represented an initial attempt to better define the Middle and Late Proterozoic Laurentia APWP (Elston and Grommé, 1974, 1984; Elston and Scott, 1973; Elston, 1986; Elston, 1989a,b,c). However, much of this work has never been published in peer-reviewed journals; some reports have only been in the form of abstracts (e.g., Elston and Grommé, 1984; Elston, 1986), or in the form of basic compilations of paleomagnetic poles used in published figures of APWPs. Unfortunately, both types of publications lack ample presentation of the raw data, in particular results of progressive demagnetization, and discussion of any field-based tests of the antiquity of the magnetizations (e.g., Elston, 1989a,b,c).

Relevant to this study, Elston and Scott (1973) published paleomagnetic results from Cardenas Basalts and lowermost strata of the Nankoweap Formation. They reported three pole positions. These were based on data from (1) two flows in the Cardenas Basalts, with a paleopole at 0.4°N, 174.6°W (10 samples), (2) five samples from the top weathered-zone of the Cardenas Basalts, with a paleopole at 49.4°N, 167.8°W and (3) six samples from the lowermost Nankoweap Formation, with a paleopole at 12.5°N, 174.4°E. None of the paleomagnetic poles reported by Elston and Scott (1973) are similar to the results presented in this study (Table 4). Although there is no obvious reason for the discrepancy between the two datasets, we offer some explanations. One is the very small number of samples and sites collected by Elston and Scott (1973). Our current analysis revealed that at least one specimen, and often several specimens, from every site within the Unkar igneous suite and the Nankoweap Formation was characterized by a spurious demagnetization response and/or yielded a randomly oriented magnetization that was not interpretable. Furthermore, within the Cardenas Basalts only 60% of the flows gave interpretable demagnetization results. Consequently, the limited sample size of the collection reported by Elston ad Scott (1973) could have led to erroneous paleomagnetic directional information and, thus, inaccurate pole positions. An additional possibility for the discrepancy is the use of more modern and sensitive methods and techniques for demagnetization in the present study compared to those used three decades ago.

The age of the UI pole is best defined by new $^{40}Ar/^{39}Ar$ results reported in this study, which provide an emplacement age of about 1090 Ma for the Unkar intrusions. Although the dated sills are located outside the area sampled for the paleomagnetic work, their age is similar to previously reported isotopic age determinations for the Cardenas Basalts and associated intrusions in the study area (Elston and Mckee, 1982; Larson et al., 1994; Timmons et al., 2001). We suggest that the newly dated sills are part of the same magmatic event that produced the mafic rocks studied for paleomagnetism and, thus, infer that the UI pole corresponds to an age of about 1090 Ma. When plotted on Laurentia's Proterozoic APWP, the UI pole is in close agreement with poles derived from other mafic intrusions, of similar age, from the southwest US (for review, see Harlan, 1993), as well as from

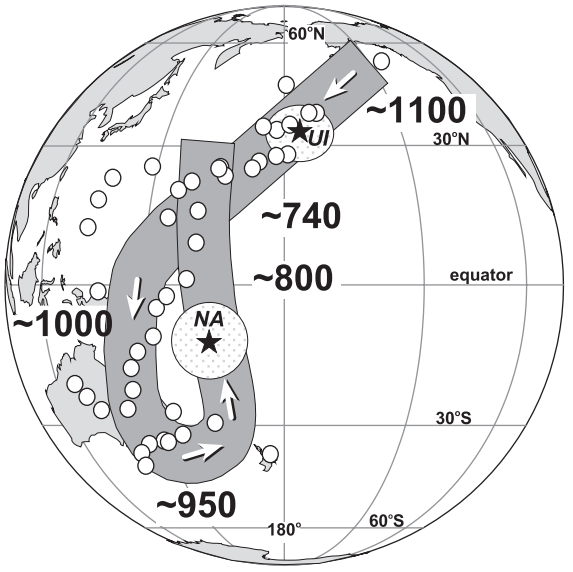


Fig. 13. Summary apparent polar wander path for Laurentia for the 1100–700 Ma time interval. Open circles are selected Keweenawan and Grenville "A poles" from Weil et al. (1998) and Warnock et al. (2000). UI, Cardenas Basalts/intrusions (Unkar Group); NA, Nankoweap Formation.

Keweenawan rocks of the midcontinent rift system (Fig. 13), which have magnetization acquisition ages of about 1090–1100 Ma (for review, see Halls and Pesonen, 1982). We note that whatever process is responsible for the disturbed $^{40}Ar/^{39}Ar$ age spectra reported by Timmons et al. (2001), which yielded total gas age estimates ranging from about 988 to 771 Ma, it does not appear to be recorded in the paleomagnetism of these rocks. The similarity in age, and paleomagnetic pole positions, of the Unkar igneous suite with other mafic suites in the southwest US (including those that intrude metamorphic rocks of the Mojave and Arizona terranes, the sedimentary rocks of the Apache and Pahrump groups and within the Salt River Canyon), the Pikes Peak Granite batholith (Smith et al., 1999) and the midcontinent rift system (van Schmus and Hinze, 1985) indicate pervasive mafic and related magmatism throughout the Laurentia craton at ca. 1100 Ma (Harlan, 1993). Heaman et al. (1992) and Howard (1991) suggest that such a continent-scale magmatic event may indicate the initiation of thermal upwelling in the mantle via multiple plumes. Hoffman (1989) speculated that this large-scale mantle upwelling was the ultimate consequence of long-term mantle insulation below a stable supercontinent. Such insulation would result in the upward deflection of lithosphere geotherms, elevated topography, and eventually large-scale melting in the mantle. Other possible explanations for such large-scale magmatism involve extension, either related to initial stages of continental break-up (Timmons et al., 2001), or to Grenville collision to the east and southeast (Lambeck, 1983; DeRito et al., 1983). Although widespread 1.1 Ga magmatism is well documented for North America, the driving force(s) and cause(s) of this event are still elusive and require further study.

The similarity between the UI pole with poles from the Keweenawan rift system of northeast North America (Fig. 13) also supports the general contention that there has been only a modest magnitude of rotation of north-central Arizona (i.e., the Colorado Plateau) with respect to the stable interior of North America during the Phanerozoic. We recognize that a comparison involving a single, reasonably well-dated pole with the track of poles defined by the Keweenawan rift system, especially given the orientation of the track with respect to the plateau, does not allow a robust estimate of the magnitude of plateau rotation. Nonetheless, the overall ensemble of paleomagnetic data from the GCSG (e.g., Elston, 1989c, 1993; Weil et al., 1999, Timmons et al., 2001; this paper) is coincident with the Keweenawan Loop and younger, Neoproterozoic results from elsewhere in North America. This observation contrasts with recent estimates of up to 10–15° of clockwise rotation of the plateau based on paleomagnetic pole comparisons between Upper Triassic strata from the southwest and the east coast of North America (e.g., Kent and Witte, 1993; Kent and Olsen, 1997; Steiner and Lucas, 2000). For a review of the Colorado Plateau rotation debate and a rigorous assessment of the maximum magnitude of plateau rotation, see Molina Garza et al. (1998).

The age of the NA pole is less well-defined than that of the UI pole. No absolute ages have been reported for the Nankoweap Formation. Previous age estimates were based on initial paleomagnetic results (Elston and Scott, 1976; Elston, 1989a,b,c). Link et al. (1993) proposed that the lower Nankoweap member is similar in age to the Cardenas Basalts, while the upper Nankoweap member would be similar in age to lower Chuar Group strata. Our new data from the Nankoweap Formation provide a paleomag-

netic pole with an estimated age of ~900–850 Ma, based on its position relative to recently obtained results from the Chuar Group (Weil et al., 1999). The inferred ~200 Ma age difference between the upper Unkar Group and the Nankoweap Formation implies that a third major unconformity exists within the Grand Canyon similar to the estimated magnitudes of time missing at the *Great* (Walcott, 1894; Noble, 1914) and *Greatest* (Noble, 1914) unconformities, already well-documented. This interpretation is similar to the proposal of Lucchitta and Hendricks (1983) that the Nankoweap Formation is considerably younger than the Cardenas Basalts, and that the unconformity between the Unkar Group and the Nankoweap Formation represents a major hiatus in the Grand Canyon sequence.

6. Conclusions

Paleomagnetic results from the Unkar igneous suite and Nankoweap Formation, GCSG, Arizona yield two primary paleomagnetic poles for Laurentia for the latest Middle and early Late Proterozoic (Fig. 13). New $^{40}Ar/^{39}Ar$ age determinations from an Unkar sill give an interpreted intrusion age of about 1090 Ma, similar to previously reported geochronologic data for the Cardenas Basalts and associated intrusions (Fig. 12). The pole position for the Unkar Group Cardenas Basalts and intrusions is in good agreement with ca. 1100 Ma paleomagnetic poles from the Keweenawan midcontinent rift deposits (Halls and Pesonen, 1982; Harlan, 1993 and references therein) and other SW Laurentia diabase bodies (Harlan, 1993 and references therein). The close agreement in age and position of the UI pole with poles derived from rift related rocks from the southwest and Keweenawan of Laurentia indicate that mafic magmatism was essentially synchronous and wide spread throughout Laurentia at ca. 1100 Ma, suggesting a large-scale continental mafic magmatic event (Howard, 1991). The pole position for the Nankoweap Formation, 40° south of that from the Unkar mafic rocks, argues for a considerably younger age of deposition, about 900 Ma, than has previously been proposed. Consequently, the ~200 Ma difference in age between the Cardenas Basalts and overlying Nankoweap Formation provides evidence for a third major unconformity within the Grand Canyon sequence.

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